

ANDREW WITTMAN

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EDUCATION

University of Virginia, Charlottesville, VA

B.S. in Mechanical Engineering

August 2022 - May 2026

- 3.93/4.0 GPA
- Relevant coursework in Mechatronics, Dynamics, Material Science, Fluid Mechanics, Advanced Mechanics, Heat Transfer
- Teaching assistant for Biomechanics

SKILLS

- **Programming:** MATLAB, Python, C++, R, Microsoft Excel, LabVIEW, microcontroller programming, Keyence VisionEditor
- **Design:** SOLIDWORKS, Autodesk Fusion 360, Ansys, Rhino, optics, soldering, laser cutting

RELEVANT EXPERIENCE

Medtronic, Minneapolis, MN

Machine Vision Engineer Intern

June 2025 - August 2025

- Conducted feasibility studies of machine vision inspection and process monitoring for automated manufacturing processes using convolutional neural networks and a classification model with 97% accuracy to save \$250,000/yr
- Designed in SOLIDWORKS a modular line scan testbench to reduce experimental setup time by 60%, employing an XYZ gantry for rough adjustment and rack and pinion fine adjustment
- Evaluated designs in reviews with team to inform criteria for testbench capabilities in sensor format, working distance, and target resolution for adjustment and imaging
- Constructed lab demonstrations of 2D area/line scan, 3D laser profiling, and 3D snapshot vision systems for Medtronic leadership

Gizmologists: University of Virginia, Charlottesville, VA

President

April 2023 - Present

- Designed and built a clock with a gantry, stepper motors, and servo motor with a stylus to write time on pressure-sensitive boards using an Arduino microcontroller and C/C++
- Ideated and constructed belt-driven core XY mechanism to move an electromagnet under a chessboard to control magnetized chess pieces in a physical match between a player and a virtual chess engine
- Developed and communicated club objectives such as training new members to improve project participation and progressing on design projects

Virginia Motorsports Formula SAE Team, Charlottesville, VA

Member: Cockpit and Controls

August 2024 - Present

- Researched and implemented thermodynamic model in Python for brake rotor temperature over duration of a simulated lap by creating estimates for heat partition between brake rotors and brake pads and estimating convection to ambient air
- Performed thermal loading studies using Ansys finite element analysis tools to assess thermal stresses on brake rotors at temperature ranges determined in Python thermodynamic model

Biomedical Engineering Design Project, Charlottesville, VA

Quantitative Pronator Drift Test For Assessment of Stroke

March 2024 - May 2024

- Collaborated with a team of five biomedical engineering students to design a device to be worn by a patient measuring downwards arm drift and wrist pronation, while returning live and summary data to administrator
- Programmed an Arduino script to calibrate sensors, read and display live data to LCD display, then calculate late-test averages for wrist pronation (roll) and downward drift (pitch)
- Developed knowledge in I2C communication and addressing to interface multiple MPU-6050 accelerometers with a TCA9548A multiplexer to retrieve data from each on separate I2C channels

LEADERSHIP ACTIVITIES

Pi Tau Sigma Mechanical Engineering Honor Society, Charlottesville, VA

President

April 2025 - Present

- Led organization of top 25% of mechanical engineering students to perform service activities to university community

Society of Hispanic Professional Engineers, Charlottesville, VA

Convention Chair

- Budgeted \$14,000 to send 20 engineering students to SHPE National Convention in Philadelphia
- Organized professional development activities for attending members